TuitionPilot 2.0: implementation report (Appendix B)

17 October 2026. Everything below runs in **SIMULATION. No real money moves.** All routes, quotes, payees and balances are fictional fixtures.

1. Repository baseline and stack

- Baseline: commit `ec77771` (v0.1.0). Python ≥ 3.10, standard library only, 39 tests, static demo site. There was no Next.js app, no database and no server. Full audit: AS-IS.md.
- Decision: build the 2.0 core as a new pure-Python package `tuitionpilot/treasury/` next to the working v0.1 engine; no Next.js rewrite before the 20 Oct deadline (ADR-001).
- Workbook conflicts and fixture choices (dates, R4, liquidity, units, caps): ADR-002.

2. Requirements completed (file paths, tests)

Module	Files	Tests / evidence	Status
Money in minor units, explicit rate direction	`treasury/currency.py`	QA-001/002/003	implemented
Route quotes: timestamp, TTL, source, data class; itemised all-in cost	`treasury/quotes.py`, `fixtures.py`	QA-001, QA-021	implemented
Deadline intelligence: business days, holidays, cut-off, time zone, latest safe arrival	`treasury/calendar.py`	QA-023	implemented
Liquidity timeline per currency, reserve, confirmed vs forecast	`treasury/liquidity.py`	QA-005, QA-006	implemented (single source currency)
Deadline-constrained optimizer: 20 candidates, hard constraints first, INFEASIBLE with reasons, baseline comparison, decision record	`treasury/optimizer.py`	QA-004, QA-005, QA-007	implemented
Tuition Shield: immutable beneficiary versions, field-level diff, trusted callback only, 48 h hold	`treasury/beneficiary.py`, `engine.parse_invoice`	QA-008, QA-009	implemented
Policy engine POL-001…POL-023, ALLOW/ASK/DENY/RETURN_EXISTING, reason codes, rule IDs, policy version, fail closed	`treasury/policy.py`	QA-010/011/012, policy tests	implemented
Exact-action approval bound to an immutable fingerprint; token hash only; recheck on redemption	`treasury/intents.py`, `engine.py`	QA-013, QA-014, expiry test	implemented
Concurrency-safe cap reservations	`treasury/limits.py`	QA-015 (20 threads)	implemented (single process)
Kill switch (user only), idempotent executor, timeout and retry	`treasury/engine.py`	QA-016, QA-017, duplicate-invoice test	implemented

Module	Files	Tests / evidence	Status
Ledger: H_n = SHA-256(canonical_json ‖ H_(n−1)), redaction, checkpoints, JSONL/CSV export, fail-closed on outage	`treasury/ledger.py`	QA-018/019/020/022	implemented
Stress and counterfactual lab	`treasury/stress.py`	`evidence/stress.json`	implemented
9 typed tools, max 6 calls per turn, user scope, tokens never returned to the model	`treasury/tools.py`	tool tests, QA-022	implemented
Flagship 10-step demo and fixed-seed replay	`treasury/flagship.py`, `__main__.py`	QA-024, `evidence/`	implemented
Demo site page “Payment plan 2.0” and a persistent SIMULATION bar on every page	`demo/template.html`	screenshots at 1280 px and 400 px, no console errors	implemented (static, recorded outputs)

The status of every QA case, policy rule and requirement is written into `docs/TuitionPilot_2_0_Backlog_with_Evidence.xlsx` (sheets QA_Test_Cases, Policy_Rules, Requirement_Map, Engine_Results, Roadmap).

3. Financial example (fixtures, assumptions visible)

Obligation: 45,000.00 CNY due 2026-09-03; university processing buffer 5 business days → latest safe arrival 2026-08-27. Planning time: Mon 2026-08-17 09:00 Asia/Almaty. Mid rate 0.01480 CNY per 1 KZT (fixture).

Route	At mid	FX spread	Variable fee	Fixed + intermediary	Known all-in (KZT)	Arrival (bd)	Result
R1 Bank wire	3,040,541	36,930	76,937	1,450	3,155,857	5–7	feasible (baseline)
R2 Fintech	3,040,541	18,353	61,178	1,030	3,121,102	2–4	feasible
R3 Account	3,040,541	12,211	45,791	620	**3,099,163**	1–3	**selected**
R4 Cheap, slow	3,040,541	9,149	30,497	350	3,080,537	7–10	rejected: arrives 2026-08-31
R5 Stale quote	3,040,541	15,279	45,837	430	3,102,087	2–4	rejected: quote 24 h old

Twenty candidates (5 routes × 4 schedules); 9 pass every hard limit. Selected: R3, one transfer, 3,099,162.83 KZT. Versus the R1 baseline on the same amount and deadline: **56,694.12 KZT (1.80%) less, SIMULATED, not a real or guaranteed saving.** Receiving-bank charges are unknown and excluded.

Stress results (`evidence/stress.json`): FX shocks change only later tranches. A 10% weaker KZT makes the four-part R2 plan breach the 500,000 KZT reserve (workbook C5). A delay of up to 5 business days keeps R3 feasible with 0 days of buffer. Sponsor funds arriving 10 days late make every plan **INFEASIBLE** with reasons. An outage on R3 moves the choice to R2 (+21,939 KZT).

Stale quotes or a changed beneficiary block every plan. Planning on 1 Aug instead makes R4 feasible and cheapest.

4. Policy, security and adversarial results

Command: `python -m pytest -q` → **74 passed** (39 v0.1 + 35 v2.0). JUnit XML: `evidence/pytest-junit.xml`.

- Critical false-allow outcomes in this suite: **0**. This is a test gate, not proof of zero real-world risk.
- Invoice prompt injection: denied (POL-018); caps, tier and kill switch unchanged; no execution.
- Changed bank details in a polite invoice: denied (POL-010); after switching to the new version: denied with a 48 h hold (POL-011); verifying from “invoice” as the contact source is rejected.
- Above the autonomous cap: ASK (POL-005, POL-009). Amount +0.01 KZT after approval: DENY (POL-021); the old approval becomes INVALIDATED.
- Replay of a used token: rejected. Expiry after 15 minutes: rejected.
- 20 parallel reservations against a 10,000,000 KZT cap: exactly 3 succeed.
- Timeout followed by two retries: 1 execution record, 1 simulated debit.
- Audit outage: DENY (POL-019); nothing reserved, nothing executed.
- Ledger: an edit in the middle is detected at the exact sequence; deleting the tail is detected only against an exported checkpoint (documented limitation).
- Raw tokens and full account numbers do not appear in the ledger or in tool outputs.
- Fixed-seed replay: identical output hash and ledger head.

5. Known limitations and TBDs

- In-memory state; locks stand in for database unique constraints and transactions. A PostgreSQL (or PocketBase after a spike) store is needed before any multi-user pilot.
- No authentication layer. `user_id` and session-user checks exist (POL-001), but there is no real identity provider.
- Fixture-only quotes and fees; quote TTL (30 min), holiday calendars, cut-offs and arrival-time ranges are TBD per provider and corridor.
- Single source currency in the liquidity book. Multi-currency wallets and converted totals need a disclosed base currency, rate and timestamp (not built yet).
- Tranche plans assume the university accepts part-payments (TBD). Later tranches are priced at today’s quote as a labelled assumption.
- The LLM planner from v0.1 is not yet wired to the 2.0 tool router; the 2.0 tools are exercised by tests and by the deterministic flagship.
- The UI is a static site showing recorded outputs. There is no Next.js app (ADR-001), and the approval-comprehension usability test with students (QA-025) has not been run.
- Hash chain: tamper-evident, not legally non-repudiable; tail deletion needs external checkpoints.
- No regulatory status of any kind is claimed. Real money requires legal, compliance and partner review.

6. Run, test, seed and replay

```
python -m pip install -e ".[test]"
python -m pytest -q                                  # 74 tests
python -m tuitionpilot.treasury flagship --out evidence   # 10-step demo + evidence files
python -m tuitionpilot.treasury replay --seed 7          # identical hashes expected
python tools/build_demo.py                            # rebuild the demo site from engine runs
```